Synthesized Feedback Report — Investing in Artificial General Intelligence

Rank: 10 of 159

Dimension scores: Innovation 73 · Methodology 68 · Relevance 82 · Rigor 80 · Overall 82

Summary

This paper develops a continuous-time model of AI infrastructure investment in which frontier laboratories choose not only when to invest and how much capacity to install, but how to split that capacity between model training (future capability building) and inference serving (current revenue generation). The model is motivated by documented capital commitments exceeding \$200 billion in 2024 alone, with projections reaching \$660–690 billion in 2026, and the \$500 billion Stargate commitment providing a concrete anchor. Demand follows a geometric Brownian motion with regime-dependent drift, switching irreversibly from a low (pre-AGI) state to a high (post-AGI) state as a Poisson event with arrival rate λ. In the low regime, competition is governed by a Tullock contest over inference capacity; in the high regime, competition shifts to a Tullock contest over training quality. A firm with total capacity K allocates fraction ϕ to training and $1-\phi$ to inference. The model incorporates Leland-style endogenous default risk, duopoly preemption following Fudenberg-Tirole and Huisman-Kort, and convex investment costs calibrated to GPU supply constraints. The paper delivers three main results: (1) an analytically characterized optimal training fraction with closed-form comparative statics (Proposition 1); (2) a faith-based survival condition (Equation 19) showing that training investment can lower the default boundary by raising the expected post-AGI continuation value through the effective revenue coefficient A_{eff} (Proposition 2); and (3) a quantitative characterization of Dario's dilemma — the asymmetric cost of miscalibrated AGI-timing beliefs — showing 26% value loss from aggressive overinvestment versus 6% from conservative underinvestment at baseline parameters. A stylized calibration to four AI lab archetypes (Anthropic-like, OpenAI-like, Google-like, xAI-like) illustrates cross-sectional implications.

Contribution Claim Assessment

The paper's primary claimed contribution is the training-inference allocation mechanism: a firm's choice of ϕ links its growth option value to its default boundary through the effective revenue coefficient A_{eff}, producing mechanisms that cannot be generated by any existing subset of real options, strategic investment, or structural credit models. This claim holds up cleanly. Existing real options models (Guo-Miao-Morellec 2005, Huisman-Kort 2015) treat capacity utilization as given. Existing structural credit models (Leland 1994) have exogenous growth options. Existing R&D race models (Loury 1979, Reinganum 1982) treat pre-breakthrough expenditure as pure cost with no concurrent revenue. The training-survival channel — where ϕ affects both the competitive position in regime H and the cash flow available to service debt in regime L — genuinely sits at the intersection of all three and cannot be derived from any subset. The paper is transparent about this, and the related-literature review is notably honest about the building blocks.

The faith-based survival result (Proposition 2) is the paper's most novel analytical contribution. The closed-form threshold $\bar{\phi}$ (Equation 19) characterizing when the A_eff-channel dominates the current cash-flow sacrifice is a genuine contribution to the structural credit literature: existing Leland-type models have exogenous growth options and cannot produce a condition under which higher training investment lowers the default boundary. The Dario's dilemma result is well-motivated and the economic mechanism is transparently explained, though the 26%/6% asymmetry is a numerical finding at specific calibration points rather than an analytically proven theorem.

The leader-follower training divergence (Proposition 3) is the paper's most overstated claim. The paper describes the leader allocating more to training than the follower as a result of preemption eliminating the contest-share cost — a clean economic story. However, this result is characterized as a "computational regularity" rather than an analytical theorem. Uniqueness of the preemption trigger is confirmed by grid search over 500 evaluation points, not analytically. The paper does not overstate this — it clearly labels Proposition 3's components as Computational, Analytical, and Numerical in Table 4 — but readers should understand that the strategic mechanism underlying the paper's most novel duopoly result lacks formal proof.

The derivative elements are substantial. The regime-switching demand structure follows Guo-Miao-Morellec (2005) closely. The default framework is standard Leland smooth-pasting. The preemption construction follows Fudenberg-Tirole and Huisman-Kort. The Tullock contest success function is a standard modeling tool. The aggregate novelty lies in the interaction structure, not in any single imported technique.

Major Comments

1. Proposition 3's key strategic result lacks an analytical proof. The leader-follower training divergence is described verbally as arising from the leader's monopoly-phase market share being insensitive to training allocation — a compelling mechanism. Yet the proof that the leader always weakly exceeds the follower's training fraction is a numerical finding verified at specific calibration points, not an analytical theorem. This is the paper's most novel strategic result, and without a proof it is unclear whether the mechanism is robust to alternative functional forms (non-Tullock contest, non-power-law scaling), asymmetric capacity choices, or off-baseline parameter values. The paper should either (a) provide an analytical sufficient condition, even approximate, under which the result holds, or (b) conduct systematic comparative statics showing how the leader-follower training gap varies across the parameter space and specifically testing alternative contest specifications beyond the Cournot robustness in Appendix E.

2. The regime-specific revenue structure is deliberately extreme. The model assumes inference revenue is zero in regime H and training value is zero in regime L. The paper acknowledges this abstraction but does not quantify how sensitive the headline Dario's dilemma magnitudes (26% vs. 6%) are to a more realistic mixed-revenue structure in which inference revenue remains large post-AGI (as is plausible — inference demand for AI services will not disappear after AGI arrives). A robustness section reporting the 26%/6% split under, say, a 50/50 or 75/25 inference-to-training revenue weighting in regime H would substantially strengthen the quantitative calibration. Without this, the specific numbers in the Dario's dilemma analysis carry limited empirical credibility.

3. The static ϕ assumption excludes the firm's most important actual decision. The paper acknowledges in Section 5.4 that firms reallocate compute on weekly timescales and that the static model overstates the initial training fraction. The two-period extension confirms the direction of bias. However, the framing of the limitation is too passive: the static ϕ assumption is not merely a tractability concession but potentially a qualitative mischaracterization of the decision problem. If firms can freely reallocate, the training-survival channel may operate differently — firms could opportunistically shift to inference during downturns, weakening the default risk tied to ϕ. The paper should clarify whether the static ϕ model provides a bound on, or an approximation to, the dynamic problem, and estimate the quantitative gap using the two-period calibration already developed.

4. Dario's dilemma reconciliation is qualitative. The paper's model shows that overinvestment carries a 6% value loss versus 26% for underinvestment — meaning overinvestment is cheaper in percentage terms. The paper then appeals to absolute dollar losses at scale (Amodei's trillion-dollar quote) to reconcile this with the verbal narrative that overinvestment is the graver risk. This reconciliation is qualitative and potentially circular: the Amodei quote refers to default risk from overinvestment, not value loss from miscalibrated beliefs, so the two quantities are not directly comparable. The paper should either (a) reframe the Dario's dilemma result to clarify that underinvestment destroys more expected value at baseline parameters while overinvestment generates more tail risk (default probability), making explicit what "asymmetric" refers to in each context, or (b) compute the Dario's dilemma cost in terms of default probability change rather than (or in addition to) percentage value loss.

5. The Leland framework is a poor fit for the primary calibration targets. The paper acknowledges that VC-backed labs are not natural users of the Leland structural default framework, which is designed for public-debt issuers. "Default" is reinterpreted as failed funding round or liquidity crisis. This reinterpretation is reasonable but consequential: VC-backed firms face discrete funding-round checkpoints rather than continuous default boundaries, have equity investors who can inject capital at distress, and have capital structures very different from the coupon-bearing debt assumed in Leland. The paper should discuss more carefully which predictions are robust to this mismatch (qualitative cross-sectional orderings likely are) and which are fragile (quantitative default boundary levels and credit spreads likely are not).

6. The small cross-section prevents formal testing. With four lab archetypes, the calibration can illustrate cross-sectional implications but cannot formally test the model's predictions. The paper is honest about this but could be more proactive about identifying near-term data sources that would discipline the calibration. AI firm credit spreads, disclosed capex-to-revenue ratios, and cloud computing contract disclosures are available for some public firms (Alphabet, Microsoft) and could provide at least partial discipline on key parameters (r, δ, γ).

Identification / Methodology Deep-Dive

What works well. Propositions 1 and 2 are analytically proven with complete, auditable proofs. The existence and uniqueness of the optimal training fraction ϕ^* is established via Inada conditions and strict concavity (second derivatives computed and signed). The $A1=0$ exactness argument in the Proposition 1 proof is careful — the result is an exact consequence of Assumption A3, not an approximation. Approximation errors in the regime-switching default reduction are quantified at below 1% at baseline parameters by comparing the unconditional A_{eff} framework against a full coupled ODE solution. Table 4 classifies every result by analytical status (Analytical, Computational regularity, Numerical finding), a degree of transparency that is unusual and commendable. Assumption 1 lists all maintained parameter conditions (A1–A4) with numerical verification that the baseline calibration satisfies them ($\beta_{L+} \approx 3.01$, $\Phi_L \approx 1.67$). The Cournot robustness in Appendix E shows that all qualitative results survive a change in competition specification, and the static ϕ bias is analyzed via a two-period extension that characterizes the direction of bias (static model overstates ϕ^*).

What is less compelling. The solution methodology is entirely standard throughout: HJB equations, ODE substitution, smooth-pasting, Fudenberg-Tirole backward induction. No new mathematical technique is introduced; the contribution is entirely in the economic structure. Proposition 3's uniqueness is established by grid search (500 points), and the leader-follower training divergence lacks an analytical sufficient condition. The Dario's dilemma asymmetry (26% vs. 6%) is a point estimate at baseline parameters; whether it holds — and in which direction — for off-baseline configurations is not shown. The revenue-inflating property of the Tullock specification under asymmetry (total industry revenue exceeds the symmetric benchmark when $y_1 \neq y_2$) could amplify preemption incentives in ways that are not separately identified from the training-survival channel; a calibration of this amplification would be useful. The calibration of λ from executive statements (Amodei 2024 implying $\lambda \approx$ 0.30–0.50; Hassabis 2025 implying $\lambda \approx$ 0.10–0.15) is stylized rather than estimated.

What to consider for robustness. The most important missing robustness check is sensitivity of the Dario's dilemma magnitudes to the regime-specific revenue structure. A sweep over mixed-revenue parameterizations (e.g., inference contributes fraction $\theta \in [0,1]$ of total revenue in regime H) would show whether 26% vs. 6% is a robust finding or an artifact of the extreme $\theta = 0$ assumption. A second priority is sensitivity of the faith-based survival threshold $\bar{\phi}$ to the Tullock parameter α and the regime-switch intensity λ, since Equation 19 applies only to the symmetric duopoly and the asymmetric case lacks a closed-form threshold. Third, the leverage comparative statics (Figure 4) should be complemented with implied credit spread estimates that can be compared against observable credit spreads for AI-adjacent public companies.

Minor Comments

- The paper uses "λ" for both the AGI arrival rate and occasionally as a shorthand in the calibration discussion — a minor but recoverable notation collision. Consider using λ_{AGI} or λ_T to distinguish.
- Figure 4 would benefit from a panel showing default probability as a function of ϕ at baseline leverage, which would directly visualize the faith-based survival mechanism more intuitively than the operating-region boundary plot.
- The verbal narrative about Dario's dilemma draws heavily on Amodei (2026) quotes, but the model's 26%/6% result is a value-loss metric, not a default probability metric. A brief clarifying sentence translating between the two concepts would reduce reader confusion.
- The A_{eff} coefficient (Equation 17) is the key quantity linking training allocation to default risk, but its economic interpretation is introduced gradually across Sections 2.3 and 2.4. A standalone paragraph early in Section 2.4 defining A_{eff} and stating its role before deriving it formally would improve readability.
- The paper's discussion of the Leland framework mismatch with VC-backed firms (Section 5.3) is appropriately candid but placed late. Given that the four calibration archetypes are all private VC-backed labs, the mismatch deserves a brief mention in Section 3 when the calibration is introduced, with a forward pointer to Section 5.3.

- The calibration in Table 2 provides training fraction estimates of $\phi \approx 0.30$–0.60 with ± 0.10 uncertainty. A sensitivity table showing how the headline quantitative results (optimal ϕ^*, default boundary X_D, Dario's dilemma magnitudes) vary across this ± 0.10 range would be a useful addition to the calibration section.
- Section 6 (conclusion) repeats material from Section 5 without adding much. Compressing the conclusion and using the space for a brief discussion of welfare implications (excessive duplication of compute investment, as in Loury 1979) would add value.
- The relation to Hackbarth, Mathews, and Robinson (2014) and Kumar and Yerramilli (2018) is noted in the related literature but not revisited when presenting the model's capital structure results. A brief comparison of the faith-based survival condition to these papers' investment-default results would sharpen the contribution.

Specific Actionable Suggestions

1. Add an analytical sufficient condition, even approximate, for Proposition 3(ii) (leader trains more than follower), or conduct systematic sensitivity analysis of the leader-follower training gap across the parameter space and flag when it reverses.
2. Compute Dario's dilemma magnitudes under mixed-regime revenue specifications ($\theta \in \{0.25, 0.50, 0.75\}$ inference revenue weight in regime H) and report them alongside the baseline $\theta = 0$ results.
3. Reframe the Dario's dilemma summary to explicitly distinguish value-loss asymmetry from default-probability asymmetry, resolving the apparent tension with the Amodei quote about bankruptcy risk.
4. Extend the static-ϕ bias analysis to report, in addition to the direction of bias, an order-of-magnitude estimate of the quantitative gap using the two-period model already developed.
5. Add a robustness table showing how the faith-based survival threshold $\bar{\phi}$ (Equation 19) varies with λ and α, since the closed form applies only to the symmetric duopoly and off-axis behavior is not characterized.
6. Add a panel to Figure 4 (or a companion figure) showing default probability as a function of training fraction ϕ, making the faith-based survival mechanism directly visible.
7. Move the brief Leland-framework mismatch caveat to Section 3 when the calibration archetypes are introduced, with a forward pointer to the Section 5.3 discussion.
8. Add a sensitivity table to Section 3 showing how optimal ϕ^*, X_D, and Dario's dilemma magnitudes vary across the ± 0.10 training-fraction uncertainty range stated in Table 2.
9. Report implied credit spreads for each archetype and compare against any available observable benchmarks (e.g., AI-sector CLO spreads, venture debt rates) to partially discipline the structural credit parameters.

10. Clarify the economic interpretation of the revenue-inflating property of the Tullock specification (Section 2.4.1, third paragraph) and discuss whether it quantitatively amplifies the preemption trigger relative to a symmetric Cournot specification.
11. Compress Section 6 and use the space to sketch welfare implications of the duopoly training arms race (total compute expenditure relative to the social optimum), connecting to the Loury (1979) excessive duplication result.
12. Add a notation table or define A_eff prominently at the start of Section 2.4 before it appears inside equations, given its central role throughout the paper.

Recommended Venue Tier

This paper is a natural candidate for field journals at the intersection of corporate finance and financial economics, and with revisions could plausibly reach a top general journal. The most natural fits are the Journal of Finance, the Review of Financial Studies, and the Journal of Financial Economics, where real options and structural credit papers with strong theoretical novelty and high-relevance empirical motivation regularly appear. The combination of timely topic, genuine analytical novelty in Propositions 1 and 2, and unusually honest handling of limitations positions the paper well for these venues. The primary obstacle to a top-journal outcome is Proposition 3's computational status — a revise-and-resubmit at JF or RFS would almost certainly require either an analytical proof or a more systematic characterization of when the leader-follower training divergence holds and when it does not. The regime-specific revenue structure concern is also likely to be raised: referees will ask for the mixed-regime robustness.

If top-general-journal submission is attempted without resolving these issues, the Review of Corporate Finance Studies (RCFS) or the Journal of Financial Intermediation represent strong second-tier field homes, particularly given the paper's direct relevance to the financing of AI infrastructure and startup capital structure. For a pure-theory contribution of this type, Management Science (Finance Department) is another plausible venue, given its receptivity to models with management implications and calibrated numerical results. The paper's policy section (Section 5.2) is underdeveloped relative to its relevance claims; expanding that section (e.g., compute governance, export controls, AI safety implications of the training arms race) would improve fit for journals interested in policy-relevant finance.

Open Questions and Extensions

- **Can the leader-follower training divergence be proven analytically?** The intuition — that monopoly-phase market share is insensitive to ϕ, reducing the marginal cost of training relative to the duopoly phase — is clean, but its validity may hinge on the Tullock functional form and the specific contest structure. Understanding what general conditions on the contest success function are sufficient for the result would determine how broadly the mechanism applies to other competitive settings.

- **How does the model change with $N > 2$ competitors?** The paper acknowledges that an N-firm extension (along the lines of Bouis, Huisman, Kort 2009) is feasible but would sacrifice closed-form tractability. A key empirical question is whether the training-survival channel intensifies or attenuates as the number of frontier labs increases — the current compute arms race involves more than two players, and qualitative predictions could differ substantially.
- **What happens when ϕ is dynamic?** The two-period extension in Section 5.4 shows the static model overstates ϕ^*, but does not characterize the dynamic optimal policy. A continuous-time model with dynamic ϕ reallocation would determine whether the training-survival channel survives dynamic adjustment, or whether firms would optimistically over-allocate to training early and rebalance toward inference as the default boundary approaches.
- **Can AGI-timing beliefs be identified from observables?** The calibration inverts observed investment behavior to implied λ values, but the mapping from λ to optimal investment is one-to-many given the other model parameters. A structural identification strategy using observable capex, leverage, and credit spread data for public AI-adjacent firms (Microsoft, Alphabet) could potentially separate λ from other parameters and produce testable predictions about cross-sectional variation in training intensity.
- **What are the welfare implications of the training arms race?** The duopoly preemption equilibrium produces earlier investment and higher training fractions than a social planner would choose — a standard preemption over-racing result amplified by the training channel. Characterizing the welfare loss (excess compute expenditure, elevated default risk) relative to the social optimum, and comparing it to regulatory benchmarks such as compute governance thresholds, would connect the model directly to current AI policy debates.